**Telecommunications Technology Solutions Inc. (TelTech)**
Corporate Tender Secretariat · Procurement & Network Engineering Division

RFP-TELTECH-FWA-2026-001 / APPENDIX C
**Author:** OLIVER TUNGOL
**Date:** October 14, 2026

# Service Level Agreements (SLAs), Network KPIs & Liquidated Damages

Operational Performance Benchmarks, Severity MTTR & Financial Penalty Schedules

| DOCUMENT NUMBER | AUTHOR | TARGET SPECTRUM | BIDDING STATUS |
|---|---|---|---|
| RFP-TELTECH-FWA-2026-001 | OLIVER TUNGOL | Band n50 (1.5 GHz, 60 MHz TDD) | Official Request for Proposal |

## 1. OBJECTIVE & GOVERNANCE PRINCIPLES

This Appendix defines the mandatory Service Level Agreements (SLAs), Key Performance Indicators (KPIs), measurement methodologies, reporting intervals, and financial remedies (Service Credits and Liquidated Damages) governing:

1. **Turnkey Deployment & Commissioning Phases (Lots 1, 2, 5)**
2. **5G Radio & Core Network Operational Performance (Lots 1, 3)**
3. **OSS/BSS Platform Reliability & Field Dispatch Operations (Lot 4)**
4. **Managed Services & First Line Maintenance Operations (Lot 6)**

Performance shall be evaluated on a calendar monthly basis using the automated performance management OSS and audited ticketing records.

## 2. PROJECT ROLLOUT & DELIVERY SLAS

### 2.1 Milestone Delivery Schedule

The Contractor shall adhere strictly to the following turnkey delivery schedule for Phase 1 (1,000 sites):

* **Milestone 1 (TSSR & Nominal Engineering Approval):** Within **45 Calendar Days** of Notice to Proceed (NTP — January 15, 2027) for all 1,000 priority sites.
* **Milestone 2 (First 200 Sites Commercial On-Air):** Within **120 Calendar Days** of Notice to Proceed (NTP).
* **Milestone 3 (Cumulative 500 Sites Commercial On-Air):** Within **210 Calendar Days** of Notice to Proceed (NTP).
* **Milestone 4 (Cumulative 1,000 Sites Commercial On-Air & Integrated):** Within **365 Calendar Days (12 Months)** of Notice to Proceed (NTP).

### 2.2 Delay Liquidated Damages (Rollout)

- If the Contractor fails to achieve Final Acceptance of a milestone by the scheduled completion date, the Operator shall deduct **Delay Liquidated Damages (DLD)** equal to **0.5% of the total milestone contract value per calendar week of delay**, up to a maximum aggregate cap of **10.0% of the total contract value**.
- Delays exceeding 60 calendar days beyond the milestone date shall entitle the Operator to terminate for default and draw down on the Contractor's Performance Bond.

## 3. RADIO ACCESS NETWORK (RAN) OPERATIONAL KPIS & SERVICE CREDITS

### 3.1 Core Radio KPIs (Calculated 24/7/365 Nationwide)

```
Radio Availability (%) = [(Total Operating Cell Hours – Total Unplanned Outage Cell Hours) / Total Operating Cell Hours] * 100
```

| KPI IDENTIFIER | METRIC DESCRIPTION | MEASUREMENT METHODOLOGY | TARGET STANDARD | MINIMUM THRESHOLD | SERVICE CREDIT PENALTY (< THRESHOLD) |
|---|---|---|---|---|---|
| **KPI-RAN-01** | **Radio Network Availability** | Monthly aggregate across all active cells | **>= 99.95%** | **99.70%** | 1.0% of monthly O&M fee per 0.1% shortfall |
| **KPI-RAN-02** | **5G Session Setup Success (CSSR)** | Initial RRC connection + 5G QoS flow setup | **>= 99.60%** | **99.00%** | 0.5% of monthly O&M fee per 0.2% shortfall |
| **KPI-RAN-03** | **5G Session Drop Rate (CDDR)** | Abnormal session drops during active data sessions | **<= 0.15%** | **0.50%** | 0.5% of monthly O&M fee per 0.1% excess |
| **KPI-RAN-04** | **Loaded Cluster Sector DL Throughput** | Busy hour average sector throughput across cluster | **>= 450 Mbps** | **380 Mbps** | Mandatory RF re-tuning within 5 days |
| **KPI-RAN-05** | **Cell-Edge DL User Speed** | 5th percentile user speed (>= 95% of test points) | **>= 50 Mbps** | **25 Mbps** | Mandatory antenna tilt optimization |
| **KPI-RAN-06** | **Cell-Edge UL User Speed** | 5th percentile user speed (>= 95% of test points) | **>= 15 Mbps** | **5 Mbps** | Uplink power control re-configuration |
| **KPI-RAN-07** | **Radio Round-Trip Latency (RTT)** | CPE to local gNodeB/UPF edge | **<= 18 ms** | **30 ms** | Audit packet buffering & backhaul |
| **KPI-RAN-08** | **Packet Loss Rate (PLR)** | User plane packet discard rate | **<= 0.01%** | **0.10%** | Transport & radio QoS re-profiling |

## 4. 5G CORE NETWORK & CLOUD PLATFORM KPIS

### 4.1 5GC High Availability Benchmarks

| KPI IDENTIFIER | METRIC DESCRIPTION | MEASUREMENT TARGET | CALCULATION INTERVAL | LIQUIDATED REMEDY |
|---|---|---|---|---|
| **KPI-5GC-01** | **Control Plane NF Availability (AMF/SMF/UDM)** | **>= 99.999%** ("Five Nines") | Monthly | $10,000 USD per hour outage exceeding threshold |
| **KPI-5GC-02** | **User Plane (UPF) Packet Availability** | **>= 99.999%** | Monthly | $10,000 USD per hour outage exceeding threshold |
| **KPI-5GC-03** | **PDU Session Establishment Success Rate** | **>= 99.80%** | 24-hr Rolling | Root-cause analysis within 12 hours |
| **KPI-5GC-04** | **CG-NAT Port Allocation Success Rate** | **>= 99.99%** | Monthly | $5,000 USD penalty per incident causing unallocated subscriber sessions exceeding 10 minutes |
| **KPI-5GC-05** | **Session-Preserving Active-Active Failover Time** | **<= 30 Seconds** (Zero session state loss; user traffic restored <= 30s) | Semi-annual Drill | Penalty of 5% annual core AMC if drill fails |
| **KPI-5GC-06** | **UPF Internal Packet Latency** | **<= 1.0 ms** | Continuous | Hardware NIC DPDK tuning |

## 5. OSS/BSS & FIELD OPERATIONS SLAS

### 5.1 Commercial & Operational Velocity Metrics

| KPI IDENTIFIER | SERVICE LEVEL DOMAIN | MANDATORY TARGET | IMPACT ON FAILURE |
|---|---|---|---|
| KPI-BSS-01 | Converged Charging System (CHF) Uptime | >= 99.99% | 2.0% monthly BSS maintenance fee deduction |
| KPI-BSS-02 | Top-Up Processing Speed (GCash / Maya) | <= 3.0 Seconds | Mandatory optimization of API gateway queues |
| KPI-OSS-01 | New Installation Dispatch Lead Time | <= 48 Hours | >= 95% of orders completed within 48h SLA |
| KPI-OSS-02 | First-Time-Right (FTR) Installation Rate | >= 96.0% | Repeat truck rolls at Contractor expense |
| KPI-OSS-03 | Churned CPE Hardware Recovery Rate | >= 70.0% | Contractor debited $55 USD per unrecovered unit below 70% |
| KPI-OSS-04 | Automated Work-Order Dispatch Engine Capacity | >= 3,500orders/day | Zero backlog accumulation during peak campaigns |

## 6. MANAGED SERVICES & FIELD INCIDENT RESTORATION SLAS

### 6.1 Incident Classification & Mean Time to Restore (MTTR)

Incidents logged by the 24/7 NOC are classified into four severity tiers:

| SEVERITY LEVEL | DEFINITION & OPERATIONAL IMPACT | RESPONSE SLA | MTTR SLA (URBAN) | MTTR SLA (RURAL) |
|---|---|---|---|---|
| Severity 1 (Critical) | Complete macro site outage; Core NF failure; >500 subscribers affected; major backhaul failure. | <= 15 Mins | <= 2.0 Hours | <= 4.0 Hours |
| Severity 2 (Major) | Single sector down; degraded throughput; loss of DC battery backup; generator fault. | <= 30 Mins | <= 4.0 Hours | <= 6.0 Hours |
| Severity 3 (Minor) | Non-service affecting hardware fault; loss of redundancy (single PSU failed, SFP degraded). | <= 1 Hour | <= 12.0 Hours | <= 24.0 Hours |
| Severity 4 (Warning/Info) | Configuration inquiry, routine audit, planned preventative maintenance scheduling. | <= 4 Hours | <= 48.0 Hours | <= 72.0 Hours |

### 6.2 Service Credits for MTTR Breaches

For every hour (or fraction thereof) that a Severity 1 incident exceeds the mandated MTTR threshold, the Operator shall deduct:

* **Urban / Suburban Sites: 0.5% of monthly site Managed Services fee per hour** of prolonged outage.
* **Rural Sites: 0.25% of monthly site Managed Services fee per hour** of prolonged outage.
* Total monthly service credit deductions for Managed Services shall be capped at **20.0% of the total monthly Managed Services fee**.

## 7. MONTHLY REPORTING & GOVERNANCE CADENCE

1. **Daily Operational Flash Report:** Generated automatically at 06:00 PHT summarizing the previous 24 hours of network availability, outages, MTTR compliance, and work orders.
2. **Weekly Engineering Review:** Joint operations meeting reviewing open trouble tickets, cluster drive tests, and rollout milestone progress.

3. **Monthly Governance & SLA True-Up:** Formal executive session reviewing all KPI metrics against SLA thresholds, finalizing service credits, and approving contractor invoices.

---

*(End of Appendix C — Service Level Agreements, Network KPIs & Liquidated Damages)*

---